

Product Requirements Document (PRD)

MedGuardian AI

Powered by Ontomorph Digital Twin Platform

Version: 1.0

Hackathon: Ontomorph Hackathon 2026

Duration: 3 days

Platform: Web (Responsive)

Team Size: 3

Executive Summary

Product Vision

MedGuardian AI is an AI-powered preventive healthcare platform built on the **Ontomorph Digital Twin Platform**. Unlike conventional symptom checkers, MedGuardian creates a **living digital health twin** for every user, enabling personalized health tracking, predictive insights, intelligent monitoring, and seamless sharing with healthcare providers.

The application combines Ontomorph's Digital Twin infrastructure, FHIR interoperability, biomarker analytics, health simulations, and provider APIs into a single intelligent healthcare assistant.

Our mission is simple:

Help people detect health problems early, understand their health better, and make informed healthcare decisions before minor issues become emergencies.

Problem Statement

Millions of people:

- Ignore symptoms until they become severe.
- Self-diagnose using unreliable internet sources.
- Lose track of medical history.
- Forget medications.
- Miss routine health checks.
- Lack access to organized personal health records.
- Have no predictive insight into their future health.

Healthcare today is reactive.

MedGuardian makes it proactive.

Solution

MedGuardian transforms every user into a **Digital Twin** using Ontomorph.

Instead of storing isolated medical records, the platform continuously builds a complete digital representation of the user's health.

Every interaction enriches the twin.

Symptoms.

Biomarkers.

Medications.

Doctor visits.

Lab reports.

Lifestyle.

Emergency events.

The result is an intelligent health companion capable of understanding the user's complete health context.

Target Audience

Primary Users

- Adults
- Students
- Families
- Elderly users
- Chronic disease patients

Secondary Users

- Doctors
 - Clinics
 - Hospitals
 - Caregivers
 - Healthcare organizations
-

Core Value Proposition

Your health shouldn't start at the hospital.

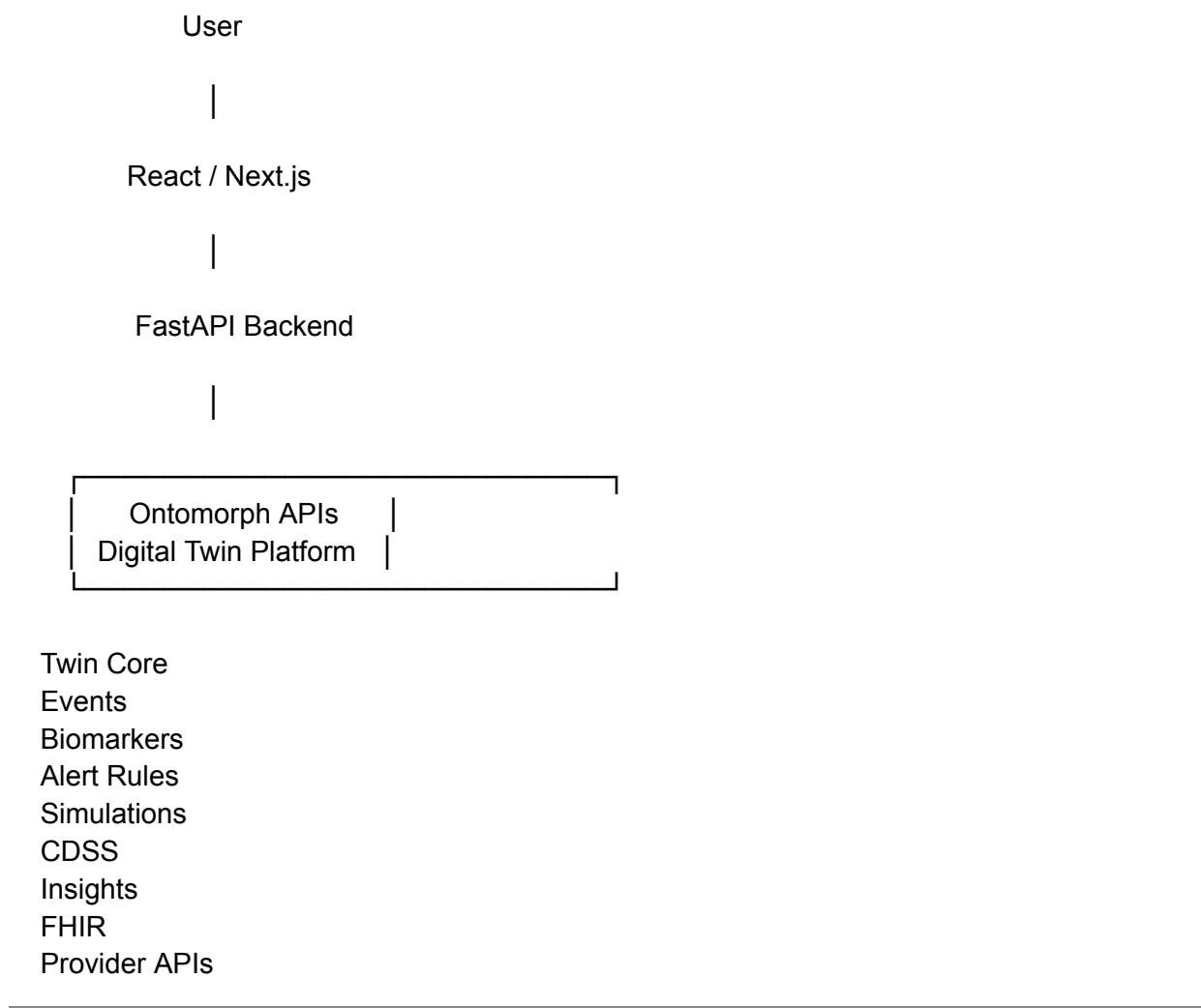
Instead of waiting until illness becomes serious,

MedGuardian continuously monitors health,

predicts risks,

and empowers better decisions.

Platform Architecture



User Journey

Step 1

User registers.



Authentication.



Create Ontomorph Digital Twin.

Step 2

User enters personal information.

- Age
- Sex
- Height
- Weight
- Medical history
- Allergies



Stored inside Ontomorph Twin.

Step 3

User reports symptoms.



AI analyzes symptoms.



Health Event created.

↓

Twin updated.

Step 4

Biomarker trends evaluated.

↓

Risk calculated.

↓

Personalized recommendations.

Step 5

Emergency detection.

↓

Alert triggered.

↓

Nearby hospital recommended.

Step 6

Doctor receives clinical summary.

↓

FHIR export.

Features

1. Digital Health Twin

Description

Every user owns a Digital Twin.

The twin becomes the single source of truth for their health.

Stores

- Demographics
 - Medical history
 - Allergies
 - Biometrics
 - Events
 - Lab results
 - Lifestyle
 - Family history
 - Genetics (future)
-

Ontomorph APIs

Twin Core

- Create Twin
 - Update Twin
 - Get Twin
 - List Twins
-

2. AI Symptom Checker

User types:

I've had chest pain for two days.

AI

- Extract symptoms
 - Ask follow-up questions
 - Suggest possible conditions
 - Assess urgency
 - Recommend next steps
-

Ontomorph Integration

Create Health Event.

Store inside Digital Twin.

Update patient history.

3. Health Timeline

Every interaction becomes part of a timeline.

Example

May 3

Fever

↓

May 10

Medication prescribed

↓

June 1

Blood pressure elevated

↓

July 5

Recovered

Ontomorph

Events API

Snapshots API

4. Biomarker Tracking

Users can monitor

- Blood sugar
- Cholesterol
- BMI
- Weight
- Blood pressure
- Oxygen saturation
- Heart rate

Charts display changes over time.

Ontomorph

Biomarker Trends API

Metrics API

5. AI Health Insights

Instead of raw numbers,

users receive explanations.

Example

Your blood pressure has increased consistently over the last month.

The AI then recommends discussing this trend with a healthcare professional.

Ontomorph

Insights API

Analytics API

6. Emergency Detection

When dangerous symptoms are detected

Example

- Stroke
- Heart attack
- Difficulty breathing

The app immediately

- Displays emergency warning
 - Shows emergency card
 - Recommends nearest hospital
 - Notifies emergency contact (optional)
-

Ontomorph

Alert Rules

Emergency Card

Events

7. Medication Assistant

Users ask

"What is Metformin?"

AI explains

- Uses
- Side effects
- Interactions
- Warnings

Medication history is saved.

Ontomorph

Events API

Clinical Mappings

8. Predictive Health Simulation

One of the biggest differentiators.

The user asks:

What happens if I continue ignoring my blood pressure?

The system simulates possible future health outcomes.

Example

Today

↓

Elevated BP

↓

3 Months

Higher cardiovascular risk

↓

1 Year

Potential hypertension complications

The app presents this as educational guidance rather than certainty.

Ontomorph

Simulations API

Provider Simulations

9. Health Risk Score

MedGuardian generates

Overall Health Score

Example

92/100

Low Risk

Contributors include:

- Biomarkers
 - Symptoms
 - Lifestyle
 - Medical history
 - Trends
-

Ontomorph

Analytics

Insights

Biomarkers

10. Clinical Summary

One-click doctor summary.

Includes

- Symptoms
 - Timeline
 - Medications
 - Biomarkers
 - Recommendations
-

Ontomorph

Provider Clinical Summary

FHIR Export

11. Hospital Finder

Maps nearby hospitals.

Displays

- Distance
 - Estimated travel time
 - Specialties
-

External APIs

Google Maps

OpenStreetMap

12. FHIR Medical Record Export

Users own their health records.

They can export

- Entire history
- Visits
- Biomarkers
- Medications

FHIR compliant.

Ontomorph

FHIR Export API

FHIR APIs

13. Temporary Doctor Access

Patient grants

Doctor access

for

30 minutes

or

24 hours

After expiration,
access is revoked automatically.

Ontomorph
Temporary Access

14. Hidden Health Records

Users can hide
Sensitive events
Examples

- Mental health
 - HIV
 - Genetic findings
-

Ontomorph
Hidden Events API

15. Secondary Findings (Future)

If genomic data exists,
the platform helps manage:

- Education acknowledgements
 - Genetic counselling referrals
 - Clinical disclosures
-

Ontomorph

Secondary Findings API

16. Voice Health Assistant

User

"I've had headaches for four days."

AI responds conversationally.

Speech APIs

-

Ontomorph

Events

Twin

Insights

17. Doctor Dashboard

Doctors can

Search patient

Review history

Clinical summaries

Biomarker trends

Risk analysis

Treatment notes

Ontomorph

Provider APIs

Provider Events

Clinical Summary

18. Personalized Preventive Recommendations

Example

You have missed your blood pressure check for three months.

Schedule one this week.

or

Your BMI has increased steadily over six months. Consider increasing physical activity and discussing this trend with a healthcare professional.

Ontomorph

Insights

Analytics

Alert Rules

19. Health Snapshots

Capture the user's health state at important moments:

- Before treatment
- After treatment
- Annual checkups
- Recovery milestones

This allows side-by-side comparisons to measure progress.

Ontomorph

Snapshots API

Screens

1. Splash Screen
 2. Login / Registration
 3. Onboarding
 4. Dashboard
 5. Digital Twin Profile
 6. AI Health Chat
 7. Symptom Analysis
 8. Health Timeline
 9. Biomarker Dashboard
 10. Health Simulation
 11. Health Risk Score
 12. Medication Assistant
 13. Emergency Screen
 14. Hospital Finder
 15. Clinical Summary
 16. Settings
-

Tech Stack

Frontend

- Next.js
- React
- Tailwind CSS
- TypeScript

Backend

- FastAPI (Python)

Database

- PostgreSQL

Authentication

- Clerk or Firebase

APIs

- Ontomorph Digital Twin Platform
- Google Maps / OpenStreetMap

Deployment

- Vercel (Frontend)
 - Railway / Render (Backend)
-

MVP (One-Week Scope)

To maximize your chances within the hackathon timeline, prioritize:

- User authentication
- Create and manage an Ontomorph Digital Twin
- AI symptom checker
- Health Events logging
- Personal health timeline
- Biomarker dashboard
- Emergency detection using Alert Rules
- AI-generated health insights
- Clinical summary generation
- Hospital finder

Features such as provider dashboards, voice interactions, genomics, and advanced simulations can be demonstrated as roadmap items if time is limited.

1. Understand the Core Philosophy of Ontomorph

This is the biggest mindset shift.

You're **not building an AI chatbot**.

You're **building an application around a patient's Digital Twin**.

Many teams will likely build:

User



AI Chatbot



Response

Instead, your architecture should look like:

User



Digital Twin



Health Events



Biomarker Trends



Insights



Simulations



Recommendations

The Digital Twin is the product's foundation—not the chat interface.

2. Read Every API Before Designing

From the documentation, these are the API groups you should fully review:

Highest Priority

- ✓ Twin Core
- ✓ Events
- ✓ Biomarker Trends
- ✓ Alert Rules
- ✓ Insights
- ✓ Simulations
- ✓ Provider
- ✓ FHIR

These APIs will likely power 90% of your application.

3. Don't Build Features Ontomorph Already Provides

Many developers waste time recreating platform capabilities.

For example:

Instead of creating your own health history database,

use:

Events API

Instead of creating your own patient profile,

use:

Twin Core

Instead of writing your own trend engine,

use:

Biomarker Trends

Instead of creating your own alert engine,

use:

Alert Rules

Leverage the platform whenever possible.

4. Use the Digital Twin Everywhere

Every feature should answer:

"How does this update or use the Digital Twin?"

For example:

Symptom Checker

User reports fever.

↓

Health Event created.

↓

Twin updated.

↓

Risk recalculated.

↓

Timeline updated.

↓

Simulation refreshed.

One action should propagate through the system.

5. Focus on One "Wow" Feature

Most hackathon winners are remembered for one standout capability.

For MedGuardian, that could be:

Predictive Health Twin

Example:

The user has:

- increasing blood pressure
- rising BMI
- worsening glucose

The system:

- analyzes trends
- evaluates alert rules
- predicts potential risks
- shows a visual health projection
- recommends preventive action

This combines several Ontomorph capabilities into one compelling experience.

6. Design Around Real User Flows

Avoid building isolated pages.

A complete flow is more impressive:

Register



Create Digital Twin



Report Symptoms



Health Event Created



Biomarker Trends Updated



Risk Evaluated



Simulation Generated



Clinical Summary Created



Share with Doctor

7. Learn FHIR Basics

Ontomorph supports FHIR (Fast Healthcare Interoperability Resources).

You don't need to master the entire standard, but you should understand:

- Patient
- Observation
- Condition
- Medication
- Encounter
- AllergyIntolerance

These are likely the main resources your app will touch.

8. Study the Provider APIs

Many teams will focus only on patients.

A simple doctor-facing view that shows:

- patient summary
- timeline
- biomarker trends
- recommendations

could make your project more complete.

9. Treat AI as an Assistant

Don't let AI make definitive diagnoses.

Instead, position it to:

- explain information
- identify patterns
- encourage medical consultation
- prioritize urgent cases
- educate users

This is safer and more credible.

10. Plan Your Demo Before Coding

A strong three-minute demo often matters more than the number of features.

For example:

1. Register a user.
2. Create their Digital Twin.
3. Report symptoms.
4. Show the Health Event being recorded.
5. Display updated biomarker trends.
6. Run a simulation.
7. Show a clinical summary.
8. Export a FHIR record.

Design your development around this story.

11. Build a Cohesive Design System

Since you're handling UI/UX, establish:

- Color palette
- Typography
- Component library
- Spacing
- Icons
- Status colors
- Accessibility guidelines

A polished interface can significantly influence judges' impressions.

12. Security and Privacy

Because you're dealing with health information:

- Use secure authentication.
 - Protect API keys.
 - Don't expose sensitive data on the client.
 - Respect patient consent, especially for provider access.
-

13. Build for Extensibility

Even if you only implement the MVP, structure the app so future features (wearables, genomics, provider tools) can be added without major redesign.

A Suggested Development Roadmap

Phase 1 – Foundation (Day 1)

- Authentication
- Create Digital Twin
- Basic dashboard
- Ontomorph API integration

Phase 2 – Core Health Features (Days 2–3)

- Symptom checker
- Health Events
- Timeline
- Biomarker trends

Phase 3 – Intelligence (Days 4–5)

- Health insights
- Alert rules
- Predictive simulation
- Clinical summary

Phase 4 – Polish (Days 6–7)

- UI refinement
 - Error handling
 - Demo data
 - Testing
 - Demo video
 - Pitch preparation
-

One More Opportunity: Build a Feature the Judges Won't Expect

Many teams will likely stop at "AI symptom checker."

You could differentiate MedGuardian by introducing a **Health Journey** experience:

- Every symptom becomes a Health Event.
- Every event updates the Digital Twin.
- Biomarker trends are visualized over time.
- Simulations estimate potential future outcomes.
- Insights recommend preventive actions.
- Patients can securely share a complete, standardized health story with a doctor.

That demonstrates not just AI, but the full lifecycle of a patient-owned Digital Twin—exactly the kind of vision Ontomorph's platform is designed to support.